

Real-Time Multimodal Intent Recognition in Mixed Reality Using Spatio-Temporal LLM

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Abstract. Recognizing user intent in Mixed Reality (MR) environments is essential for HCI and assistive technologies, where human intent drives adaptive systems. However, it remains a challenging open problem due to the multimodal and highly dynamic nature of such interactions. We present SeeIntent — a novel end-to-end real-time multimodal emotionally-aware intent recognition system for egocentric user perspective, which fuses object-level visual context, emotional state, and spatial focus, derived using Facial Action Coding System, head rotation and gaze descriptors streamed from the MR headset. The aggregated state is encoded into a rolling temporal history and submitted to a locally-deployed open-weight Large Language Model (LLM) backend for action-intent generation. We introduce a temporal context-window architecture, termed the Brain, which maintains a history of multimodal state vectors and conditions LLM inference based on historical context. Our framework offers a replicable, dataset-agnostic methodology for the cognitive analysis of user behavior in First Person Vision research. Five LLM backend configurations of the proposed system are evaluated on EPIC-KITCHENS, a large-scale benchmark for egocentric activity understanding. To validate the proposed egocentric emotion context, we introduce a novel annotated pilot dataset collected under naturalistic, in-the-wild conditions, providing ground-truth labels for scene-level objects, fixation targets, egocentric emotional state and intended action. Results show that while the higher-parameter models achieve the highest semantic matching accuracy, smaller model variants offer competitive performance at substantially lower inference latency (under 300 ms), making them better suited for real-time interactive deployment.

Keywords: Mixed Reality · Egocentric Behavior · Intent Recognition · Action Recognition · Large Language Models · Eye Tracking · Object Detection · Emotion Classification

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1 Introduction

Mixed Reality (MR) systems are rapidly evolving from passive display technologies into active, context-aware assistants capable of understanding and anticipating user needs [3]. A key enabler of proactive MR assistance is accurate, low-latency *intent recognition*: determining what action a user is about to perform, based on the available contextual information. Achieving this requires the system to simultaneously interpret what the user sees, how they feel, and where their attention is directed — signals that are heterogeneous in modality, asynchronous in arrival, and temporally correlated in meaning.

Prior work on intent and action prediction has largely focused on unimodal cues: egocentric vision [7, 13], physiological signals [22], or gaze [1]. Multimodal fusion approaches have emerged more recently but typically operate offline on pre-segmented clips rather than in real-time [19]. Locally-deployable Large Language Models (LLMs) allow us to use natural language to reason over real-time multimodal data without the need for task-specific classifiers.

In this work, we present a modular pipeline that realises this vision for MR headsets as a proof-of-concept. Our contributions are as follows:

- A **multimodal state vector** design that encodes per-frame visual detections (via an Object Detection Module), emotional state (via an Emotion Classifier Module over facial blendshapes), and head-rotation gaze into a unified representation.
- A **temporal context-window architecture** (*the Brain*) that maintains a rolling history of state vectors and structures an LLM prompt that captures recent interaction dynamics.
- A **validation framework** leveraging the standardized, large-scale EPIC-KITCHENS dataset [7], supplemented by an initial pilot study on annotated egocentric headset sequences to demonstrate real-world end-to-end pipeline feasibility.
- **Comparative evaluation** of five open-weight LLM configurations across BLEU-1, ROUGE-L, verb/noun accuracy, hallucination rate, and inference latency, yielding actionable architectural recommendations for real-time MR deployment.

2 Related Work

We discuss state-of-the-art methods in egocentric intent recognition, gaze estimation in XR, usage of affective states in XR, and end with a note on usage of LLMs for contextualizing the multimodal input streams.

2.1 Egocentric Action & Intent Recognition

Egocentric video understanding has advanced significantly with large-scale datasets such as EPIC-KITCHENS [7] and Ego4D [13]. Transformer-based architectures [19,

28] now achieve strong performance on third-person action recognition, and recent work has extended these to anticipation tasks that predict the *next* action from a short observation window [11, 12]. However, these approaches are computationally heavy and require offline processing, making them unsuitable for real-time MR deployment. Our architecture trades long-range predictive depth for computationally lightweight, real-time inference suitable for wearable deployment.

2.2 Gaze and Attention in XR

TODO: [8] Gaze is a strong predictor of imminent interaction [20, 16]. Modern commercial mixed-reality headsets natively expose per-frame head rotation quaternions and eye-tracking vectors, enabling fine-grained attention tracking without external hardware. Several studies have fused gaze with scene understanding for object selection in VR [4], but the integration of gaze into intent-prediction pipelines that concurrently incorporate emotion and language remains largely unexplored.

2.3 Affective Computing in VR

TODO: [18], [26] Facial Action Coding System (FACS)-derived blendshapes provide a compact, hardware-agnostic representation of facial expressions [9]. Public datasets, such as *emoji-hero-vr-db* [10], offer labeled blendshape recordings from VR users, enabling emotion classifiers to be trained without external cameras. While prior work has explored emotion recognition from raw blendshapes [23], the field has seldom integrated the resulting affective states into downstream intent prediction systems.

2.4 LLMs for Contextual Reasoning

Instruction-tuned LLMs have demonstrated strong zero-shot reasoning over structured context [5, 24, 27]. Recent systems exploit LLMs as planners or intent interpreters in embodied AI settings [2, 17], but these typically operate on clean, pre-parsed inputs. Integrating noisy, high-frequency sensor streams into LLM prompts—while strictly managing latency—remains an open challenge. Recent advances in local inference frameworks make sub-second LLM calls feasible on commodity hardware, yet a systematic evaluation of open-weight models for structured intent generation in MR has not previously been reported.

2.5 Multimodal Fusion Architectures

Late-fusion strategies that aggregate unimodal predictions have proven effective when modalities are asynchronous [19]. Our system adopts a text-serialization approach: each modality is converted into a natural-language token (emotion label, object list, gaze direction), concatenated into a structured temporal prompt,

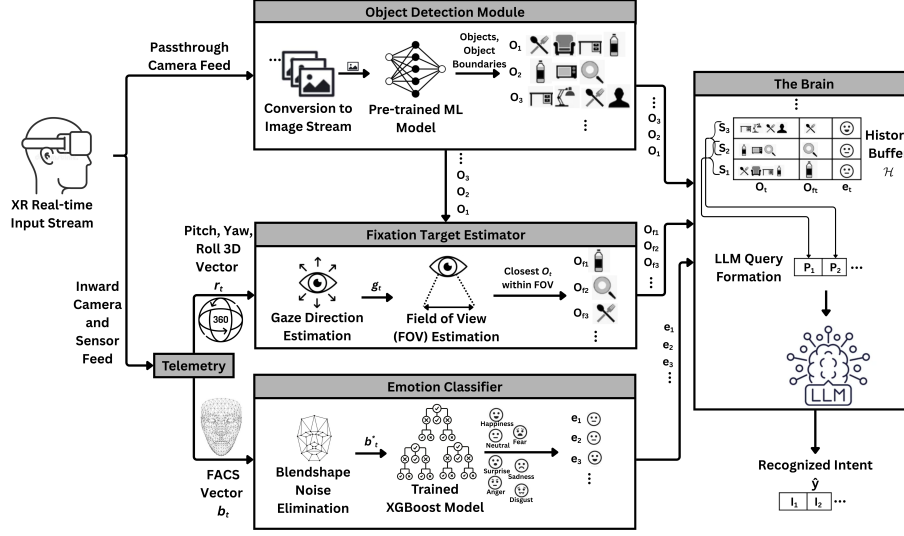


Fig. 1: Overall system architecture of the proposed system. The system considers various camera and sensor feeds as input. The different modules of the pipeline interact with each other, to give the egocentric user’s intent as output.

and reasoned over by the LLM. This architecture bypasses the need for computationally expensive cross-modal alignment layers while directly exploiting the LLM’s pre-trained commonsense knowledge.

3 Methodology

We leverage the various camera feeds and sensor inputs from commercially available XR headsets, for the task of intent recognition. Figure 1 gives an overview of the proposed system. At each time step, the system captures one frame from the MR casting window, and processes it using object detection, fixation target and emotion classification modules in parallel. The combined observation is encoded into a state vector, which is appended to a rolling history, while simultaneously triggering an asynchronous LLM-query for intent recognition. In the following sections, we look at each module’s design in detail.

3.1 Telemetry & Pose Tracking

Telemetry aids in time-synchronizing the multimodal data. In our system, in order to capture user’s movements and facial expressions in real time, we recommend a threaded UDP server to listen for JSON packets coming from a real-time 3D engine client running on the commercial MR headset. UDP is specifically chosen over TCP to eliminate handshake overhead and prevent head-of-line blocking [21], prioritizing real-time state updates and low latency over guaranteed packet delivery.

Rotational tracking of the head, given as a 3D vector of pitch, yaw and roll the headset’s cameras is computed for each timestep. In addition, the inward camera also exposes the *Facial Expression Activation Code (FACS)* values of each muscle, which is then used to compute the FACS vector.

At each timestep t , from the eye and lower-face tracking camera feed, the computed payload contains:

1. Rotation vector: A 3-DOF head rotation vector $\mathbf{r}_t \in \mathbb{R}^3$, which tracks pitch, yaw, and roll in degrees.
2. FACS vector: A 63-dimensional *Facial Expression Activation Code (FACS)* vector $\mathbf{b}_t \in [0, 1]^{63}$. These blendshape values track specific facial muscle movements (like dropping the jaw or raising an eyebrow) on a scale from 0 (neutral) to 1 (fully flexed).

3.2 Facial Emotion Recognition (FER) Model

We propose a novel integration of the egocentric user’s emotion into the traditional intent recognition pipeline [11, 8], which utilizes multimodal data consisting of gaze and scene perception. While recent efforts have shown that egocentric user’s emotion reveals user behavior and personality [18], we look into the direction of intent recognition using emotion as an input modality. While this research area is nascent, [26] and available information is access-controlled, we make an effort to use freely available public dataset and repositories to propose suitable models, that fits the device’s hardware constraints. Since inward camera’s feed would be access-restricted due to privacy policies in most headset, we leverage the XR headset inward camera’s ability to provide FACS feed — a floating-point vector of different facial muscle movements, rated from 0 to 1, provided by the module defined in Section 3.1. Due to the resource impositions of locally running ML models, we allocate GPU resources for the object detection model (of Section 3.3), while CPU resources could be allocated to the FACS-code based emotion detection module.

From various testing, we decided to build an *Extreme Gradient Boosting (XGBoost)* model [6], that would be pre-trained on a suitable FER dataset that would use FACS coding. XGBoost was selected because it demonstrates high efficiency on tabular data and can easily learn the complex patterns between different facial expressions without being computationally heavy [14].

Feature Selection The 63-dimensional FACS vector is computationally intensive to the already resource-constrained XR headset, and hence we propose a feature selection step, to eliminate noisy data and any blendshape data that is occluded due to the camera position. Using feature importance, we arrive at a subset $b_t^* \in b_t$ which consists of FACS blendshapes that ranks high in the feature importance list. While this step does incur additional latency, we hypothesize that the XGBoost model would perform faster with the smaller subset b_t^* , than with the whole FACS vector.

To determine the optimal subset b_t^* , we employ a two-tiered feature selection strategy utilizing XGBoost’s Information Gain metric. High-dimensional facial tracking data often contains overlapping signals, particularly between visually similar expressions such as Fear and Surprise, which can confuse a standard multi-class model. To mitigate this, our selection process operates in two distinct phases. First, an exploratory global classifier is trained across all emotion classes to extract the top 20 most influential blendshapes, capturing the primary variations necessary for generalized classification. Second, to address the specific discriminatory challenge between the most closely related classes, we isolate the data for Fear and Surprise and train a targeted binary classifier. This extracts an additional top 10 features specifically optimized to disambiguate these two expressions.

The final selected subset b_t^* is constructed from the union of these two feature sets, eliminating duplicates. This hybrid approach ensures the model retains both the macro-level features required for general emotion classification and the micro-level nuances necessary for edge cases, while substantially reducing the computational footprint required for real-time XR headset processing.

Our model then takes the b_t^* reliable blendshapes and classifies them into one of seven emotions defined by FACS [9]:

$$e_t = \text{XGB}(\mathbf{b}_t^*) \in \{\text{Fear, Happiness, Anger, Disgust, Neutral, Sadness, Surprise}\}.$$

DONE: Details on feature importance method used.

3.3 Object Detection Module

To facilitate real-time scene understanding, a lightweight object detection model is deployed locally and executed on available GPU hardware. The module receives sequential frames from the egocentric passthrough camera feed and produces, a set of object class labels $\mathcal{O}_t = \{o_1, o_2, \dots, o_k\}$ for each timestep t , corresponding to entities present in the observed scene. A propagation latency of approximately 20–40 ms is expected by the passthrough camera pipeline for most commercially available XR headsets. However, GPU-accelerated inference ensures that model execution latency remains well within the bounds required for real-time deployment.

3.4 Fixation Target Estimator

The module uses a combination of the scene-level object array (i.e. the output of the module described in Section 3.3, the objects’ bounding boxes and depth, and head rotation sensor output (in Section 3.1) to predict the probable gaze target.

We derive a discrete gaze direction g_t from the yaw and pitch components of $\mathbf{r}_t$, obtained from telemetry data (Section 3.1):

$$g_t = \text{encode}(\text{pitch}_t, \text{yaw}_t) \in \{\text{Centre, Left, Right, Up, Down}\}.$$

Algorithm 1: Brain Prompt Construction

Input: History $\mathcal{H}$, window size w
Output: LLM intent prediction $\hat{y}_t$

```

1  $P \leftarrow$  "You are an intent predictor in a Mixed Reality scene.";
2 for  $i = t - w$  to  $t$  do
3    $P \leftarrow P +$ 
   f"Step {i}: Emotion={ $e_i$ }, Objects={ $\mathcal{O}_i$ }, Fixation Target={ $o_{f_t}$ }"
4 end
5  $P \leftarrow P +$  "Predict the next action as a verb-noun phrase.";
6  $\hat{y}_t \leftarrow \text{LLM}(P)$ ;
```

Since foveated rendering [15] is used in XR devices, the "Centre" of the world-view is considered as the primary gaze vector, with allowed thresholds for visual field as per literature. Thresholds are $\pm 15^\circ$ for horizontal axis and $\pm 10^\circ$ for vertical axis, which reflects natural head-movement granularity in MR tasks.

From the scene-level objects $\mathcal{O}_t$, we derive the fixation target as $o_f \in \mathcal{O}_t$ whose bounding box intersects with the visual field. In case multiple objects fall within the field of view, the object with the lowest depth i.e. directly in front of the user, is designated as the fixation target.

3.5 Intent Recognition Module (The Brain)

The *Brain* module maintains a fixed-length history buffer $\mathcal{H} = [\mathbf{s}_{t-w}, \dots, \mathbf{s}_t]$ of w state vectors, where each vector is defined as:

$$\mathbf{s}_t = (e_t, \mathcal{O}_t, o_{f_t}).$$

When an LLM query is triggered, the Brain serialises $\mathcal{H}$ into a structured natural-language prompt of the form shown in Algorithm 1. The prompt instructs the model to predict the user's immediate next action as a single verb-noun phrase, drawing on both the current multimodal state and the recent trajectory.

Inference is dispatched asynchronously to either a local or cloud-based Large Language Model (LLM) backend, preventing the prediction loop from blocking frame capture.

3.6 System Integration

In most XR headset configurations, the frame rate for passthrough camera could be 60–90 *Hz*, with a display of similar refresh rate, the eye tracking camera would run at lower frame rate, around 30-60 fps. We also consider this system for casting to a remote monitoring device, which may also potentially have lower refresh rates. We thus run the main orchestration loop at the frame rate of the casting window and eye tracking module (approximately 30 fps), which is also moderated by the telemetry module. LLM queries are rate-limited to avoid saturating the inference backend; by default a new query is issued every $w = 5$ frames once

the history buffer is full. All modules communicate via shared Python objects protected by threading locks.

3.7 Dataset for Evaluation

Currently, there is no dataset available that contains egocentric user emotion, gaze information, scene-level objects and user action/intent. Hence, in order to validate the full multimodal pipeline, we conducted a small-scale pilot study, recording four video segments from a Meta Quest Pro headset worn by participants performing naturalistic daily tasks. We use this dataset as a proof-of-concept to evaluate our system.

Annotation The dataset contains four short one-minute recordings of the two primary authors performing everyday desk-based activities. Ground-truth action labels were manually annotated by the authors, specifying the intent as a verb-noun action. Each video was temporally segmented based on the duration of distinct atomic actions (e.g., "grabbing a notebook", "cutting the paper"). For each temporal segment, annotators provided a standardized verb-noun pair (e.g., verb: 'grab', noun: 'notebook'), the duration in seconds, and a self-annotated predominant facial expression selected from the FACS categories (e.g., Neutral, Happiness, Surprise).

Unlike other intent recognition evaluation datasets [7, ?], full multimodal state vectors (emotion and objects) are provided for each segment. Consequently, each discrete time window t is represented by a unified ground-truth state vector:

$$\mathbf{v}_t = [\text{emotion}_t, \text{duration}_t, \text{verb}_t, \text{noun}_t]$$

To facilitate further research and ensure the reproducibility of our baseline evaluation, the complete annotated dataset, including the JSON-formatted ground-truth state vectors, is publicly accessible in the `dataset` folder of our project's GitHub repository.

DONE: More details on the annotation and vector representation.

DONE: Mention that it is given in the dataset folder in our GitHub repo.

3.8 Evaluation Metrics

We evaluate intent recognition quality using four metrics:

- **BLEU-1:** Unigram precision of the predicted action phrase against the ground-truth annotation.
- **ROUGE-L:** Longest common subsequence recall, capturing partial structural overlap.
- **Verb Match / Noun Match:** Binary accuracy of the primary verb and primary noun in the prediction, extracted by NLTK POS-tagging.

- **Hallucination Rate:** Fraction of predictions that generate nouns not strictly present in the detected object set $\mathcal{O}_t$. *Note:* Because YOLOv11n utilizes a highly constrained vocabulary (e.g., outputting “cup” instead of “mug”), this metric serves as a strict upper bound. It captures both true confabulations and valid semantic inferences that simply diverge from YOLO’s rigid taxonomy.
- **Latency:** Wall-clock time (seconds) from prompt dispatch to response receipt.

TODO: More on evaluation metrics

4 Experiments and Results

We implement our proposed system by empirically testing various ML models for object detection, emotion classification, and various light-weight LLM models. We report on suitable evaluation metrics at each step. We also report metrics on intent recognition on two available datasets, namely, the EPIC-KITCHENS [7], an egocentric video dataset of kitchen activities, and our own in-the-wild dataset that contains everyday desk-based activities. The code for the system pipeline, and our pilot dataset is shared in our GitHub repository, which will be shared once paper is accepted for publication.

4.1 System Implementation and Sub-System Validation

To instantiate our proposed modular pipeline, we deployed YOLOv11n [25] for object detection and our custom 22-feature XGBoost model — trained on the *emoji-hero-vr-db* dataset [10] — for emotion classification, running local inference against data streams from a Meta Quest Pro headset.

Before evaluating the end-to-end intent prediction pipeline, we validated the optimized XGBoost emotion classifier offline, for the pilot dataset. Evaluated strictly on the 22 selected features, the model achieved an overall test accuracy of **78.84%** across the six target emotions. The model demonstrated high reliability in detecting "Happiness" (F1-score: 0.86) and "Neutral" states (F1-score: 0.83). The most significant point of confusion remained "Fear" (F1-score: 0.60), which the hardware frequently misclassified as "Surprise" due to their shared upper-face actuations (e.g., widened eyes and raised brows). Additionally, because the downward-facing cameras frequently struggle to differentiate between 'Anger' and 'Disgust' (since both rely on lower-face muscles that are hard to see), we decided to merge them into a single "Hostile" category to boost overall accuracy. Our optimized model takes the 22 reliable blendshapes and classifies them into one of six emotions:

$$e_t = \text{XGB}(\mathbf{b}_t^*) \in \{\text{Fear, Happiness, Hostile, Neutral, Sadness, Surprise}\}.$$

When our system starts up, it loads our saved model, the label encoder, and the list of our 22 specific features. During the main application loop, whenever

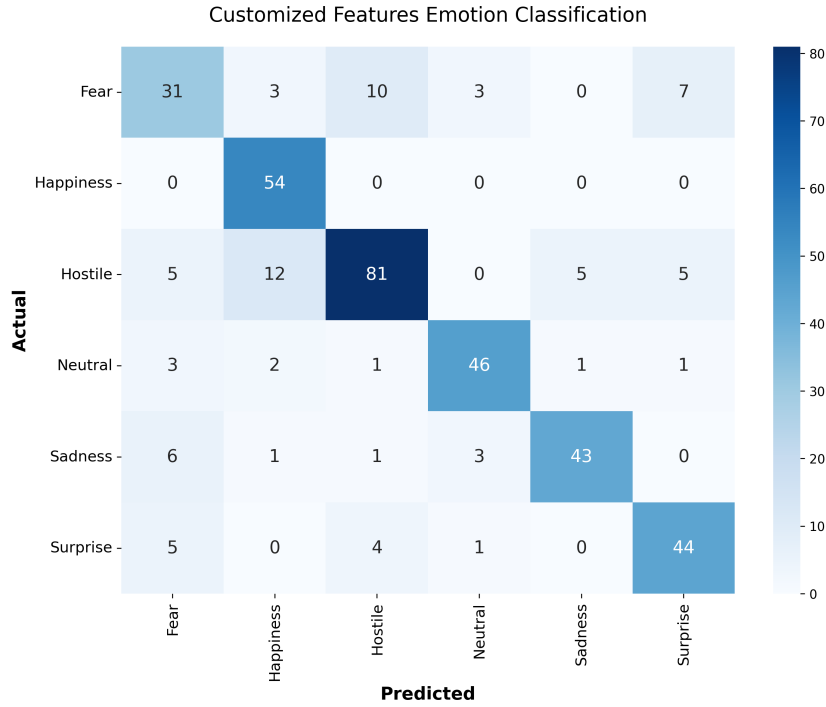


Fig. 2: Confusion Matrix for the optimized 22-feature XGBoost emotion classifier. The matrix highlights high accuracy in detecting Happiness, Neutral, and Hostile states, while revealing the hardware-induced overlap between Fear and Surprise.

a 63-dimensional vector comes in from the headset, we slice out just the 22 features we need and pass them to the XGBoost classifier. Because the model is so small, this classification takes roughly 1–2 milliseconds on the CPU, allowing us to predict emotions without slowing down the frame capture or the LLM context builder. Overall, an accuracy of nearly 79% provided a sufficiently reliable and computationally efficient baseline for the downstream LLM context window. Figure 2 shows the confusion matrix of actual vs. predicted emotions for our pilot dataset.

4.2 Intent Recognition Evaluation

All local LLMs used for the Intent Prediction Module are served via Ollama on a workstation with a single NVIDIA GPU. No fine-tuning is applied; all models are evaluated zero-shot.

4.3 Evaluation I: EPIC-KITCHENS Dataset

We sample 17 participant videos from the EPIC-KITCHENS-100 [7] validation split. At each annotation midpoint, three frames are extracted and processed

Table 1: EPIC-KITCHENS evaluation — aggregate results (mean $\pm$ std). Best in **bold**; highlighted model has lowest hallucination.

Model	BLEU-1	ROUGE-L	Verb%	Noun%	Halluc. $\downarrow$	Latency (s)
llama3.2	0.005	0.014	0.000	0.005	0.80	0.17
mistral	0.012	0.017	0.002	0.015	0.70	0.28
gemma3:4b	0.010	0.016	0.002	0.013	0.78	2.56
phi4-mini	0.011	0.017	0.003	0.012	0.63	2.01
qwen2.5:3b	0.003	0.003	0.000	0.007	0.68	1.93

Table 2: In-the-Wild Pilot Study — aggregate results (mean $\pm$ std). Best in **bold**.

Model	BLEU-1	ROUGE-L	Verb%	Noun%	Latency (s)
llama3	0.067	0.076	0.003	0.003	2.37
mistral	0.058	0.073	0.000	0.000	2.50
gemma3:4b	0.077	0.092	0.000	0.000	2.50

by YOLOv11n. The resulting object state is passed to the Brain with an empty emotion state, as no headset telemetry is available for this dataset. We evaluate five LLMs: **llama3.2**, **mistral**, **gemma3:4b**, **phi4-mini**, and **qwen2.5:3b**.

Table 1 and Figure ?? summarize the aggregate metrics. All models achieve modest BLEU-1 and ROUGE-L scores (below 0.025), reflecting the inherent difficulty of predicting fine-grained kitchen actions from object lists alone. **mistral** leads on BLEU-1 (≈ 0.012) and shows the broadest coverage on the radar chart (Figure 3). Hallucination rates are uniformly high (0.63–0.80), with **phi4-mini** achieving the lowest rate (≈ 0.63), suggesting it is the most grounded in the visual context despite moderate overlap scores.

Regarding response times, Table 1 reveals a clear two-tier split. **llama3.2** and **mistral** respond in under 300 ms (median ≈ 0.17 s and 0.28 s, respectively). Conversely, **gemma3:4b**, **phi4-mini**, and **qwen2.5:3b** require approximately 2.0–2.6 s per query—a magnitude difference with significant implications for real-time deployment.

4.4 Evaluation II: In-the-Wild Pilot Study

All three models achieve substantially higher scores on the pilot dataset (BLEU-1 up to 0.08; ROUGE-L up to 0.10) compared to EPIC-KITCHENS, confirming that emotion and gaze signals provide meaningful context. As shown in the radar chart (Figure 4), **gemma3:4b** dominates on all NLP overlap metrics, notably peaking at a ROUGE-L of 0.55 on one highly predictable segment. However, verb and noun exact-match scores remain near zero for all models, indicating that predicted phrases are semantically related but lexically diverse.

As shown in Table 2, latency was uniformly higher on this benchmark (≈ 2.3 –2.5 s median) compared to the fastest EPIC-KITCHENS models. This overhead is directly attributable to the richer, longer prompts generated from the full multimodal state histories.

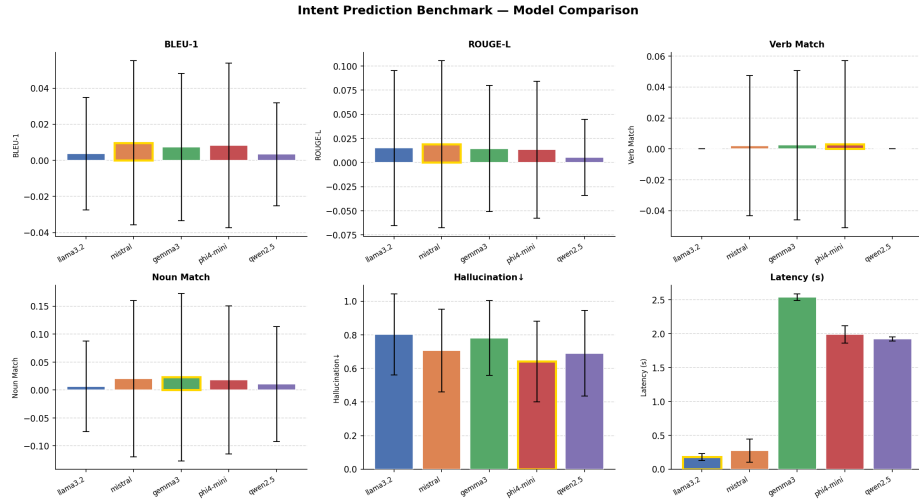


Fig. 3: EPIC-KITCHENS radar chart.

4.5 Comparison Across Evaluation Settings

Comparing both datasets reveals a *semantic bottleneck* introduced by the object detection layer. YOLOv11n maps the visual space to a highly constrained, generic vocabulary. The LLMs lack the visual granularity required to exactly match specific ground-truth annotations (e.g., predicting “use cup” instead of “wash glass”). This strictly penalizes n-gram overlap scores, even when predictions are semantically plausible.

While the in-the-wild pilot study yielded higher absolute scores than the EPIC-KITCHENS evaluation, the limited sample size precludes definitive claims regarding the absolute superiority of multimodal fusion. Rather, the pilot successfully validates the operational viability of the pipeline, proving that concurrent telemetry, vision, and LLM orchestration can execute reliably in real time.

Model parameter count does not strictly correlate with zero-shot spatial reasoning. The 7B-parameter **mistral** model outperformed larger architectures on several EPIC-KITCHENS metrics while responding nearly $10\times$ faster than **gemma3:4b**. However, uniformly high hallucination rates across all models indicate that unconstrained LLMs frequently confabulate actions beyond the observed object context. This strongly motivates replacing the YOLO layer with a high-speed open-vocabulary detector or implementing constrained decoding strategies in future work.

5 Discussion

Based on the evaluation results, we observe nuanced results, which are discussed in the following sections.

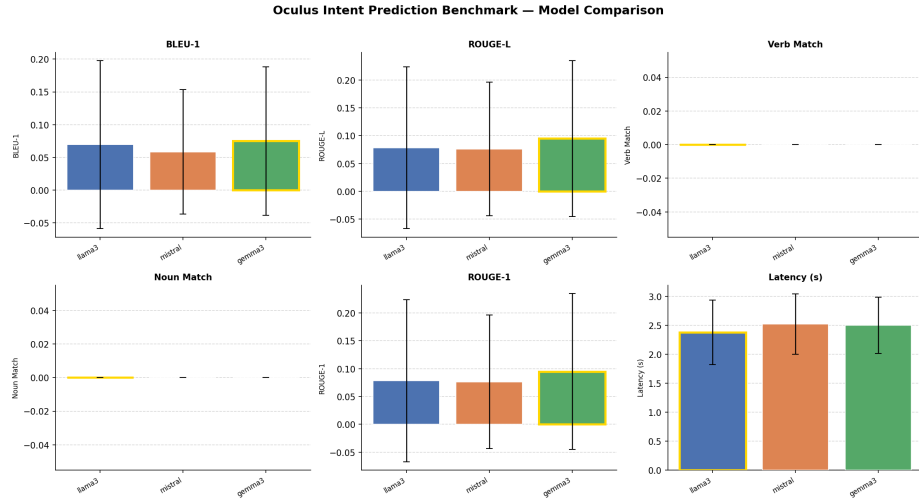


Fig. 4: Pilot Study radar chart.

Fig. 5: Model Radar Comparison — all metrics normalised to [0,1].

5.1 Module Selection Strategy

By designing the architecture as a modular pipeline, we were able to independently select optimal models for each task. The choice of YOLOv11n for object detection and XGBoost for emotion classification ensured that the vision and telemetry layers remained computationally lightweight, reserving maximum GPU resources for the LLM Intent Prediction module.

5.2 Multimodal Signals are Necessary but Not Sufficient

The improvement from the vision-only EPIC-KITCHENS evaluation to the fully multimodal Pilot Study is substantial in absolute terms, yet overall scores remain low across all NLP metrics. This indicates that while each additional modality contributes useful context, the overall task—predicting a specific verb-noun action phrase from noisy, unstructured sensor inputs—requires further advances in grounding and prompt engineering. Incorporating explicit scene-graph representations or object-relation tuples into the prompt may bridge this semantic gap.

5.3 Latency–Accuracy Trade-off

Our results reveal a clear Pareto frontier: `llama3.2` and `mistral` occupy the low-latency, moderate-accuracy region, while `gemma3:4b` achieves the highest accuracy at the cost of nearly $10\times$ longer response times. For real-time MR applications where user actions unfold on the scale of 1–2 seconds, sub-300 ms

latency is a strict practical requirement, making `llama3.2` and `mistral` the pragmatic choices. Speculative decoding or quantized model variants could potentially close the latency gap for larger architectures.

5.4 Context Deviation vs. True Hallucination

All models exhibit a strict hallucination rate above 60%, meaning the majority of predictions reference nouns not explicitly provided by the YOLO vision layer. However, qualitative review suggests this is heavily conflated with the semantic bottleneck described above. For example, if the vision layer detects a “cup” but the LLM naturally predicts the action “wash mug”, the rigid metric flags this as a hallucination. While true confabulations (inventing actions entirely unsupported by the scene) remain a safety concern for MR assistants, a significant portion of this 60% represents the LLM attempting to compensate for the detector’s limited vocabulary. Future iterations must employ semantic similarity thresholds (e.g., cosine similarity of word embeddings) rather than strict string-matching to isolate true hallucinations.

5.5 Zero-Shot Performance Ceiling

Verb and noun match accuracies remain near zero across both evaluation domains, suggesting that zero-shot LLMs cannot reliably reproduce the specific lexical choices used in ground-truth annotations. Fine-tuning on a small set of annotated MR interaction pairs would likely yield large gains on exact-match metrics while concurrently reducing the hallucination rate.

5.6 Limitations

The system currently relies on Windows-specific APIs for MR frame capture, limiting cross-platform portability. Additionally, while the XGBoost emotion classifier was rigorously validated offline, its robustness to dynamic real-world factors during live deployment—such as physical HMD slippage or diverse facial morphologies—requires further investigation. Finally, the in-the-wild pilot study comprises recordings from a limited number of participants and environments; broader, large-scale evaluation is needed to fully assess pipeline generalization.

6 Conclusions

We have presented a real-time multimodal intent prediction system for Mixed Reality environments that fuses YOLO-based visual perception, facial blend-shape emotion classification, and head-rotation gaze encoding into a temporally-structured LLM prompt. Our evaluation strategy—encompassing the standardized EPIC-KITCHENS egocentric dataset and an in-the-wild pilot study of annotated MR recordings—provides a rigorous analysis of five open-weight LLMs across accuracy, context deviation, and latency dimensions.

Key findings indicate that: (i) multimodal fusion (vision, emotion, and gaze) yields roughly $5\times$ higher NLP-overlap scores compared to vision-only baselines; (ii) **gemma3:4b** achieves the highest BLEU-1 and ROUGE-L scores across both evaluation domains; (iii) **llama3.2** and **mistral** offer competitive predictive capability at sub-300 ms latency, making them pragmatic choices for real-time MR applications; and (iv) all models exhibit high strict hallucination rates ($>60\%$), heavily underscoring the semantic bottleneck introduced by rigid object-detection vocabularies.

Future work will focus on (a) fine-tuning lightweight LLMs on domain-specific MR interaction datasets, (b) integrating a retrieval-augmented action ontology and open-vocabulary detectors to suppress context deviation, (c) extending the gaze module to leverage per-eye vergence data for precise 3D spatial grounding, and (d) conducting a comprehensive user study to assess how accurate intent prediction translates to perceived utility in MR assistants.

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